

Weite Pi

Curriculum Vitae

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Research Interest

Algebraic geometry and related fields

Employment

2025–Present Postdoctoral researcher, University of Bonn

Education

2020–2025 Ph.D. in Mathematics, Yale University

Advisor: Prof. Junliang Shen

2020–2021 Research assistant, Shanghai Center for Mathematical Sciences

2016–2020 B.S. in Mathematics, Fudan University

Publications and Preprints

1. On the Chern filtration for the moduli of bundles on curves (joint with W. Lim and M. Moreira), [arXiv:2410.24008](https://arxiv.org/abs/2410.24008).
2. Cohomological stabilization, perverse filtrations, and refined BPS invariants for del Pezzo surfaces (joint with J. Shen, F. Si, and F. Zhang), [arXiv:2406.10004](https://arxiv.org/abs/2406.10004).
3. Cohomology rings of the moduli of one-dimensional sheaves on the projective plane (joint with Y. Kononov, W. Lim, M. Moreira), [arXiv:2403.06277](https://arxiv.org/abs/2403.06277).
4. Cohomological χ -dependence of ring structure for the moduli of one-dimensional sheaves on $\mathbb{P}^2$ (joint with W. Lim, M. Moreira), *Forum Math. Sigma*. 12 (2024) No. e47, 17 pp.
5. Perverse filtrations, Chern filtrations, and refined BPS invariants for local $\mathbb{P}^2$ (joint with Y. Kononov, J. Shen), *Adv. Math.* 433 (2023) 109294, 29 pp.
6. Generators for the cohomology ring of the moduli of one-dimensional sheaves on $\mathbb{P}^2$ (joint with J. Shen), *Algebr. Geom.* 10 (4) (2023) 504–520.

Invited Talks

- Feb 2026 “ $P = C$ ” and χ -independence phenomena for moduli of 1-dimensional sheaves
Harvard–MIT Algebraic Geometry Seminar, MIT
- Oct 2025 “ $P = C$ ” and χ -independence phenomena for moduli of 1-dimensional sheaves
Algebraic Geometry and Moduli Seminar, ETH Zürich
- Oct 2025 Mumford relations and the cohomology of Le Potier moduli spaces
Seminar Algebraic Geometry, Max Planck Institute for Mathematics

- Aug 2025 “Perverse = Chern” phenomenon for moduli spaces of 1-dimensional sheaves
Yonsei University, two lectures
- Mar 2025 Perverse filtrations, BPS invariants, and moduli of one-dimensional sheaves
Algebraic Geometry and Commutative Algebra Seminar, University of Kansas
- Jan 2025 On the Chern filtration for certain moduli spaces of sheaves
Seminar on Geometry and Arithmetic, UCSB
- Nov 2024 Tautological classes, perverse filtrations, and moduli of one-dimensional sheaves
Math-Physics Seminar, University of Pennsylvania
- Oct 2024 Tautological classes, perverse filtrations, and moduli of one-dimensional sheaves
Algebraic Geometry Seminar, University of Michigan
- May 2024 Mumford relations, Virasoro representations, and moduli of one-dimensional sheaves
Algebraic Geometry Seminar, Academia Sinica
- Apr 2024 Cohomology rings of the moduli of one-dimensional sheaves on the projective plane
Algebraic Geometry Seminar, Stanford University
- Oct 2023 Moduli of one-dimensional sheaves on $\mathbb{P}^2$: cohomology and beyond
Algebraic Geometry Seminar, Stony Brook University
- Sep 2023 Moduli of one-dimensional sheaves on $\mathbb{P}^2$: from cohomology to enumerative geometry
Boston Algebraic Geometry Day, MIT
- Apr 2023 On the cohomology of the moduli of one-dimensional sheaves on $\mathbb{P}^2$
AMS New England Graduate Student Conference, Brown University
- Apr 2023 Refined BPS invariants for local $\mathbb{P}^2$ from the moduli of one-dimensional sheaves
Short talk and poster session, University of Pennsylvania
- Mar 2023 A very brief introduction to perverse sheaves
Yale visiting day for prospective Ph.D. students
- Mar 2023 Moduli of one-dimensional sheaves on $\mathbb{P}^2$: cohomology, perversity, and BPS invariants
Geometry, Symmetry and Physics Seminar, Yale University
- Jan 2023 Moduli of one-dimensional sheaves on $\mathbb{P}^2$: cohomology, perversity, and BPS invariants
Algebraic Geometry Seminar, UCSD
- Oct 2022 Refined BPS invariants from the cohomology of moduli of one-dimensional sheaves
Algebraic Geometry and Gromov–Witten Theory Seminar, University of Bonn
- Oct 2022 Moduli of one-dimensional sheaves on $\mathbb{P}^2$: cohomology, perversity, and BPS invariants
Algebraic Geometry and Moduli Seminar, ETH Zürich

Contributed Talks

- Jan 2026 Some recent progress on phantoms
Preprint Seminar, University of Bonn
- Dec 2024 Tautological classes, perverse filtrations, and moduli of one-dimensional sheaves
Algebraic Geometry Seminar, Shanghai Center for Mathematical Sciences
- Nov 2024 On the “Perverse = Chern” phenomenon for certain moduli spaces of sheaves
Graduate Student Seminar, Yale University
- Jun 2024 Cohomology rings of the moduli of one-dimensional sheaves on the projective plane
Poster session, Moduli Stacks and Enumerative Geometry, Isaac Newton Institute

- Mar 2024 On a compact analogue of the $P = W$ conjecture
Short research talk, AGNES @ Boston College
- Nov 2023 Three faces of the moduli space of vector bundles on algebraic curves
Graduate Student Seminar, Yale University
- Oct 2023 Moduli of sheaves, Mumford relations, and Virasoro operators
Short research talk, AGNES @ UPenn
- Aug 2023 Moduli of 1-dimensional sheaves on $\mathbb{P}^2$: from cohomology to enumerative geometry
Algebraic Geometry Seminar, Shanghai Center for Mathematical Sciences
- Jul 2023 Moduli of 1-dimensional sheaves on $\mathbb{P}^2$: from cohomology to Gopakumar–Vafa theory
Poster session, Derived FRG Conference, Imperial College London
- Jun 2023 Mumford relations for the moduli of one-dimensional sheaves on $\mathbb{P}^2$
Poster session, Arithmetic, Birational Geometry, and Moduli spaces, Brown University
- Apr 2023 The cohomology rings for $M_{5,1}$ and $M_{5,2}$ are different
Lightning talk, Jane Street Graduate Research Fellowship Workshop
- Feb 2023 Curve counting invariants for local $\mathbb{P}^2$ from the moduli of one-dimensional sheaves
Graduate Student Seminar, Yale University
- Nov 2022 Perverse filtrations, Chern filtrations, and refined BPS invariants for local $\mathbb{P}^2$
Poster session, AGNES @ UMass Amherst
- Sep 2022 From curves to sheaves: a tale of two moduli spaces
Graduate Student Seminar, Yale University
- May 2022 On the cohomology ring of the moduli space of one-dimensional sheaves on $\mathbb{P}^2$
Poster session, AGNES @ Rutgers
- Nov 2020 Lubin–Tate theory and totally ramified extension of local fields
Graduate Student Seminar, Yale University

Academic Experiences

- Mar 2026 Homological Algebra in Algebraic Geometry and Representation Theory, Salerno
- Feb 2026 Two-week research visit to MIT; Host: Miguel Moreira
- Dec 2025 Conférence finale de l'ANR FanoHK, Université de Toulouse
- Oct 2025 Hodge Theory and Algebraic Cycles, Clay Research Workshop
- Jul 2025 Summer Research Institute in Algebraic Geometry, Colorado State University
- May 2025 Hodge Theory and Algebraic Geometry, Stony Brook University
- Mar 2025 Georgia Algebraic Geometry Symposium, University of Georgia
- Feb 2025 Hyperkähler Varieties, Derived Categories, and Moduli Spaces, Columbia University
- Nov 2024 Algebraic Geometry Northeastern Series, Dartmouth College
- Oct 2024 Boston Algebraic Geometry Day, Harvard University
- Oct 2024 The $P = W$ Conjecture in Non Abelian Hodge Theory, Clay Research Workshop
- Jul 2024 Simons Math Summer Workshop: Moduli, Stony Brook University
- Jun 2024 Moduli Stacks and Enumerative Geometry, Isaac Newton Institute
- Apr 2024 Geometry and Representation Theory around the $P = W$ Conjecture, Oberwolfach
- Mar 2024 Algebraic Geometry Northeastern Series, Boston College

Jan 2024 Winter School on New Applications of Mixed Hodge Modules, Stony Brook University
 Nov 2023 Geometric Representation Theory and Moduli Spaces, UNC Chapel Hill
 Oct 2023 Algebraic Geometry Northeastern Series, University of Pennsylvania
 Sep 2023 Boston Algebraic Geometry Day, MIT
 Jul 2023 AGNES Summer School on Intersection Theory on Moduli Spaces, Brown University
 Jul 2023 Derived Categories, Moduli Spaces, and Counting Invariants, Imperial College London
 Jun 2023 Conference on 100 Years of Noetherian Rings, Institute for Advanced Study
 Jun 2023 Arithmetic, Birational Geometry, and Moduli Spaces, Brown University
 Apr 2023 Algebraic Geometry Northeastern Series, Stony Brook University
 Apr 2023 Categorical Methods in Moduli Theory Workshop, University of Pennsylvania
 Spring 2023 Reading group on Intersection Theory (organizer), Yale University
 Nov 2022 Algebraic Geometry Northeastern Series, UMass Amherst
 Oct 2022 Two-week research visit to ETH Zürich and the University of Bonn
 Fall 2022 Reading group on Algebraic Geometry (organizer), Yale University
 Aug 2022 AGNES Summer School on Higher Dimensional Moduli, Brown University
 Aug 2022 Derived Categories, Moduli Spaces, and Hyperkähler Varieties, University of Michigan
 May 2022 Algebraic Geometry Northeastern Series, Rutgers University
 Fall 2021 Graduate Student Seminar (organizer), Yale University
 2020/2021 Algebraic Geometry Summer School, Shanghai Center for Mathematical Sciences
 Aug 2019 “Algebra and Number Theory” Summer School, Chinese Academy of Sciences

Awards and Honors

May 2025 Department Dissertation Prize, Yale University

Teaching

Spring 2021 MATH 120: Calculus of Functions of Several Variables, Teaching Fellow
 — 2022 MATH 225: Linear Algebra, Teaching Fellow
 — 2023 MATH 120: Calculus of Functions of Several Variables, Primary Instructor
 — 2024 MATH 115: Calculus of Functions of One Variable II, Primary Instructor
 — 2025 MATH 120: Calculus of Functions of Several Variables, Primary Instructor
 2022–2025 Certificate of College Mathematics Teaching Preparation, Yale University

Service

Referee for IMRN, Asian J. Math., Adv. Math.